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Mapping nine decades of research integrity studies (1935–2024): A scientometric analysis

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ABSTRACT

Background: Research integrity is fundamental to responsible research practice. Despite attention, the intellectual structure and evolution of this field remains underexplored. This study maps the knowledge landscape of research integrity, identifying key themes, contributions, and trends.

Methods: A scientometric analysis was conducted on 6,895 records from Web of Science and Scopus (1935–2024). CiteSpace facilitated network analysis, including co-authorship, keyword co-occurrence, and co-citation patterns, while burst detection identified topics.

Results: Research integrity studies have grown significantly since the 1980s, with interdisciplinary collaboration. Keyword and co-citation analyses reveal a shift from early discussions on scientific misconduct to concerns such as open science, AI ethics, and research governance. A collaboration network has emerged, with leading contributions from North America, Europe, and Asia.

Conclusions: Research integrity has matured into an interdisciplinary field, reaching academic consensus with growing integration of policies, regulations, and technology. Future research is expected to focus on AI's role in research integrity. Key areas of concern include algorithmic bias, automation ethics, and implications for scholarly publishing. Open science and transparency will remain central, particularly in addressing data fabrication, paper mills, and predatory publishing. Institutional policies will continue evolving, embedding integrity principles into governance and public engagement initiatives.

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Research integrity;
responsible research;
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scientometric analysis;
knowledge map

1. Introduction

In 1970, Summerlin, a clinical researcher at Stanford University, announced a skin-grafting technique purportedly to circumvent transplant rejection, a claim initially celebrated as a major breakthrough in transplantation research (Cope 1973; Woolf 1988). However, the failure of independent

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scholars to replicate his experiments, coupled with mounting skepticism, cast serious doubts on the credibility of his findings, eventually leading to allegations of research misconduct. This controversy culminated in widespread media attention, with The New York Times likening it to a “medical Watergate” and the Los Angeles Times describing it as “one of the most notable examples of fraudulent scientific research” (Brody 1974; Steinbrook 1987; TIME 1974; Weissmann 2006).

The prevalence of research fraud in the 1970s and 1980s contributed to the development of the concept of research integrity, despite the fact that the two represent opposite sides of the same coin. Several high-profile cases, including the five major instances of research misconduct reported in the United States between 1979 and 1981 (Broad and Wade 1982, 232–34), the controversy over credit for the discovery of HIV in the mid-1980s, and concerns regarding data reliability in David Baltimore’s research group in 1986, heightened awareness of research misconduct and underscored the limitations of existing oversight mechanisms (Löppönen and Vuorio 2013). These events led to significant institutional reforms, including the establishment of dedicated research oversight bodies and stricter policies on scientific misconduct. More importantly, they shifted the discourse from perceiving misconduct as isolated acts of deception to recognizing the need for a systemic framework to promote research integrity. Notably, in 1987, the National Science Foundation (NSF) introduced its own research integrity guidelines, marking an important step toward formalizing policies to address research misconduct (Price 1994).

This development facilitated a growing scholarly understanding of research integrity, leading to the articulation of shared principles, norms, and beliefs that fostered responsible research practices (Pascal 1999a). Against this backdrop, research on research integrity began to emerge, with scholars increasingly focusing on institutional structures, particularly the role of institutional leadership in promoting research integrity (Gunsalus 1993) and the responsibilities of research integrity offices at the time (Pascal 1999b; Price 1998). These early studies contributed to the rapid expansion and diffusion of research integrity discussions into fields such as nursing, biomedical science, and clinical medicine (Demiroglu 2000; Friedman 1991; Lutz 1999).

Since the beginning of this century, research integrity has become a field of multidisciplinary cross-research. More complex forms of research misconduct, such as gift authorship (Manton and English 2006), paper mills (Hackett and Kelly 2020), and predatory journals (Beall 2016), have prompted scholars to focus on the impact of systems such as research evaluation (Hayer et al. 2013), academic publishing (Rifai et al. 2019), open science (Haven et al. 2022), and higher education (Abdi et al. 2021; Shephard et al. 2015) on research integrity, driving the expansion and maturation of this field. After more than 50 years of exploration, the field of research integrity has accumulated a substantial body of literature that not only has

mapped its current state and trends but also has offered insights into its developmental patterns.

However, despite being a focal point of academic attention, the internal knowledge structure and evolutionary trajectory of the research integrity field remain unclear. By tracing the knowledge production process, this study maps the intellectual landscape of research integrity and identifies emerging research hotspots, providing valuable insights for future research. Given the rapid and evolving nature of this field, this study makes several contributions:

- (1) It provides the first comprehensive analysis spanning nearly 90 years of literature, systematically revealing the intellectual evolution and structural development of the research integrity domain.
- (2) It integrates scientometric analysis with network visualization techniques to identify key contributors, influential works, and dominant research clusters, thereby elucidating the underlying knowledge diffusion patterns and interdisciplinary connections shaping the field.
- (3) It employs CiteSpace to detect knowledge bursts across different periods, offering a nuanced understanding of the distribution of active research topics in various stages of the field's evolution.

2. Literature review and research questions

By reviewing the fundamental concepts and theories of scientometric analysis, this study aims to understand how this method can contribute to mapping the knowledge landscape in the field of research integrity. Furthermore, a review of existing scientometric studies in this domain helps identify potential research gaps, which in turn guides the formulation of research questions.

2.1. *Scientometric analysis of knowledge evolution*

Nalimov and Mulchenko (1971, 2) first defined scientometrics as “quantitative methods for studying the evolution of science as an informational process.” Scientometric analysis employs quantitative approaches to examine the production, dissemination, and impact of scientific knowledge, treating knowledge as a dynamic system of interactions. One of the key objectives of scientometric analysis is to understand the identity of a discipline or research domain (Serenko et al. 2010). While knowledge structures across various fields continue to expand and diversify, it remains essential to periodically conduct retrospective analyses of a discipline or domain to address fundamental questions about its development and trajectory (Holsapple 2008).

Unlike bibliometrics, which primarily focuses on statistical patterns in academic publications, scientometrics encompasses a broader range of analyses. It not only examines bibliometric indicators but also investigates research collaboration networks, thematic evolution, and shifts in research priorities (Granovsky 2001). These aspects align with Eugene Garfield's concept of "historiographs," which map connections between key papers to reconstruct the intellectual lineage of significant discoveries (Garfield, Sher, and Torpie 1964). This perspective led to the development of co-citation analysis, which has employed techniques such as factor analysis, multidimensional scaling, and clustering to identify and visualize highly related networks of papers – thereby defining research domains and frontiers (Small 1973). Additionally, co-word analysis examines term pairings in titles, abstracts, or keywords, drawing on Callon and Latour's Actor-Network Theory (ANT) to capture conceptual linkages (Callon et al. 1983). Advances in computational algorithms and mapping techniques have further enabled the visualization of high-dimensional datasets, enhancing our ability to interpret research landscapes (De Nooy, Mrvar, and Batagelj 2005).

Beyond its methodological scope, scientometrics also investigates research indicators and their quantitative behaviors, research policy, and the organization of research institutions (Mingers and Leydesdorff 2015). In sum, scientometric approaches allow us to systematically map the intellectual trajectory of research integrity, review its accumulated knowledge base, and analyze the evolution of citation networks, collaboration patterns, and emerging research trends. As a methodology grounded in the science of science, scientometrics provides empirical insights into the development, evaluation, and forecasting of research integrity studies, offering a comprehensive methodological approach for understanding its dynamic transformations.

2.2. Scientometric studies in the field of research integrity

A number of studies have employed scientometric and bibliometric methods to review the research integrity domain. For example, a recent study (Nguyen and Tuamsuk 2024) used data from the Scopus database to compile publications related to research integrity from 2004 to 2023, analyzing dimensions such as author collaboration networks, research topics, and key journals. However, this study did not investigate the evolution of the underlying knowledge structure or the emergence of critical concepts, thereby limiting its ability to fully capture the dynamics of knowledge flow within the field.

Other investigations have focused on research integrity within specific contexts. For instance, one bibliometric analysis examined themes in the realm of higher education – addressing issues such as contract cheating, academic collusion, misconduct in online education, and other forms of scientific misconduct (Ali and Aboelmaged 2020). Similarly, another study conducted a bibliometric analysis

within the context of artificial intelligence and identified a gap in the literature regarding how AI might enhance academic integrity (Rodrigues et al. 2024). Although these context-specific studies have provided valuable insights into particular subthemes of research integrity, their narrow focus precludes a comprehensive depiction of the field's overall intellectual landscape.

In addition to scientometric and bibliometric reviews, some researchers have employed systematic review methodologies to examine literature on topics associated with research integrity. These systematic reviews have covered areas such as research funding (Roje et al. 2022), organizational culture (Vidak et al. 2021), responsible research practices (Field et al. 2024), and academic integrity (Macfarlane, Zhang, and Pun 2012). Furthermore, a study that analyzed empirical research within the research integrity literature found that although many empirical articles attribute deviations from best research practices to systemic issues, the proposed measures to promote integrity tend to focus on individual researchers' awareness and compliance rather than on systemic reforms (Aubert Bonn and Pinxten 2019). Despite the valuable contributions of these studies, their focus on specific topics or research types – and the absence of a holistic mapping of the field's knowledge structure – prevents them from fully delineating the intellectual genealogy of research integrity. In response to these limitations, the present study addresses the following research questions:

RQ1: Research Output and Collaboration Structure: What trends can be observed in the research output related to research integrity, and what structural patterns are revealed by the collaboration networks among authors, the distribution of national outputs and institutional partnerships, and the characteristics of key publishing venues?

RQ2: Terminology and Keyword Co-occurrence and Burst Detection Analysis: How do the contributions of specific keywords and terminologies illuminate the evolution of research topics in the field of research integrity? In particular, which terms emerge as dominant or transformative at different developmental stages, and what does this indicate about the flow and diffusion of knowledge within the field?

RQ3: Citation Co-occurrence and Knowledge Evolution: What insights does citation co-occurrence analysis provide regarding the structural configuration of the research integrity literature? Specifically, how does the co-citation network evolve over time, and which clusters or influential publications signal significant bursts of impact and potential future research frontiers?

By addressing these questions, this study provides a knowledge map of the research integrity field, delineating trends in research output, identifying

structural patterns in collaborations and institutional partnerships, and elucidating the evolution of research topics and intellectual structure. These insights enhance our understanding of the domain and inform future research.

3. Materials and methods

This study employs a scientometric analysis rather than a conventional bibliometric analysis, systematically examining both the structural dynamics and intellectual evolution of the research integrity field.

3.1. Data collection

3.1.1. Database selection

To ensure comprehensive data collection, both the core collection of the Web of Science (WoS) and the Scopus database were used. These two databases are widely regarded as among the most comprehensive literature databases worldwide, offering extensive coverage, rigorous inclusion criteria, and relatively complete metadata (Falagas et al. 2008). Databases such as PubMed and ERIC were not considered due to inconsistencies in their metadata formats compared to those of the aforementioned databases, which can lead to failures and errors during data format conversion in scientometric analysis.

3.1.2. Selection of search terms

The selection of search terms is critical in scientometric analysis. Drawing on established practices in previous studies (Aubert Bonn and Pinxten 2019, 339), the following terms were employed: “research misconduct,” “research integrity,” “responsible research,” “scientific integrity,” “scientific misconduct,” “scientific fraud,” and “research fraud” (see Figure 1).

The decision not to include “academic integrity” and “academic misconduct” as search terms is based on two specific considerations. First, these terms are typically used in a broad sense, encompassing topics such as classroom behavior, student plagiarism, and higher education policies – subjects that fall outside the primary focus on research practices. Their inclusion could therefore lead to heterogeneous search results. Second, although academic integrity and research integrity share similar underlying values (Krásničan et al. 2024), their contexts differ. The term “academic integrity” is predominantly associated with student behavior in higher education (Macfarlane, Zhang, and Pun 2012, 340), whereas “research integrity” is used to denote the conduct of researchers (Helgesson and Bülow 2021). Including both terms might conflate literature from distinct contexts, thereby reducing the specificity and comparability of the search results.

To enhance dataset precision and relevance, narrowly defined, context-specific search terms were used, ensuring a more homogeneous selection of literature. While this study aims to compile a relevant and representative dataset, the selection of search terms may introduce inherent biases. Measures are implemented to mitigate this bias by ensuring that data collection is conducted openly and transparently, and the limitations of the approach are clearly communicated.

3.1.3. Data cleaning

As shown in Figure 1, on 28 January 2025, searches of titles, abstracts, and keywords were conducted in both the WoS and Scopus databases. The WoS search returned 6,115 records, and the Scopus search returned 18,286 records. Subsequently, the following records were excluded: (1) To ensure

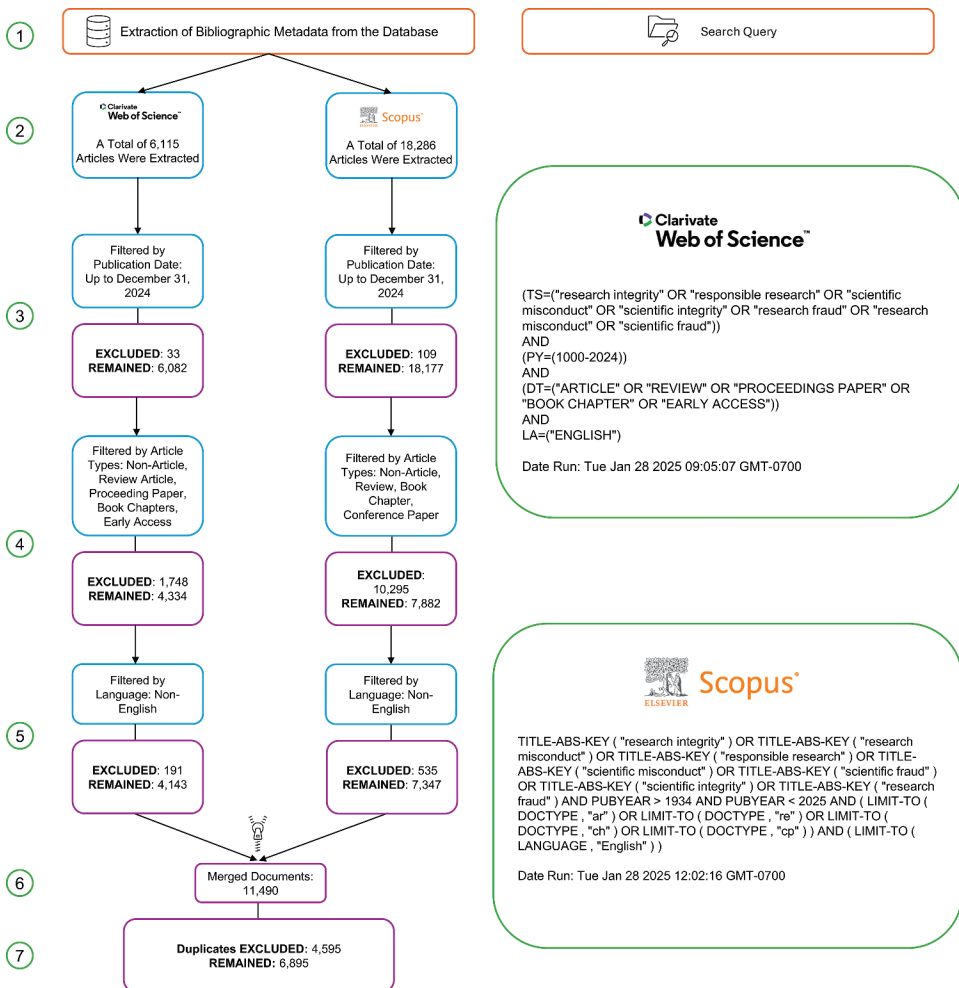


Figure 1. Flow diagram of the search strategy.

complete annual data coverage, only records up to and including 2024 were included in this study. As the search was conducted in January 2025, records from 2025 were excluded since the year was still in progress and would provide incomplete data. (2) Documents categorized as editorials and commentaries, which – despite their significant contributions to the development of the research integrity field – lack essential metadata such as abstracts and keywords, thereby limiting subsequent analyses. (3) Document types such as news items, notes, book reviews, corrections, and retractions, as these are not included as primary research sources in this study. (4) Non-English language publications, due to limitations in the CiteSpace software; including multilingual records could lead to data loading failures.

Next, the data format conversion function in CiteSpace was used to convert all records retrieved from Scopus into the WoS format. The two datasets were then merged to yield a total of 11,490 records. Finally, the automatic deduplication function in CiteSpace removed 4,595 duplicate records, resulting in a final dataset of 6,895 records for analysis.

3.2. Data analysis

The data were analyzed using CiteSpace 6.4. Multiple node types were selected for analysis, including co-authorship, institutions, countries, keywords, terms, cited references, and cited journals. A time slicing approach was employed, with each slice representing a 10-year period to capture long-term trends. Parameter settings were iteratively refined based on the visual quality of the network diagrams; the specific parameter values used for each figure are displayed in the upper left corner of the respective images, ensuring transparency and reproducibility. Keyword clustering was the primary analytical method, enabling the identification of dominant research themes and their evolution over time.¹

Considering the large size of the dataset, Latent Semantic Indexing (LSI) was selected as the clustering algorithm. LSI is particularly effective for identifying dominant research themes and their evolution over time by reducing dimensionality and filtering out noise in large-scale text data (Sehra, Singh, and Rai 2017). This choice ensures efficient computation and reliable clustering results, making it well-suited for uncovering key topics and trends in the dataset.

In CiteSpace, the Tree Ring History visualization was used to represent the temporal evolution of key nodes in the network. Node size in this visualization is determined by citation frequency, meaning that documents, authors, countries, institutions,² terms, or keywords³ that have been cited more frequently appear as larger nodes (Li and Chen 2022, 107). The outer rings of the nodes indicate different time periods, with variations in color and thickness reflecting changes in

citation activity. Thicker outer rings suggest periods of increased attention, highlighting research trends and influential works.

Co-citation analysis identifies pairs of documents that are cited together in the same source document, reflecting intellectual connections between works (Small 1973, 267). A co-occurrence network represents the frequency of paired data within a given set, such as keywords, institutions, or authors, providing insights into thematic structures and collaboration patterns.

To evaluate the structural properties of the network, modularity and clustering quality metrics were employed. The modularity score (Q) quantifies the density of connections within modules or communities, offering a measure of how well a network is divided into distinct clusters. The Q score ranges from 0 to 1, with values exceeding 0.3 indicating a significant clustering structure. Higher modularity values suggest a well-structured network with strongly defined communities (Sabé et al. 2023).

Another important metric for assessing clustering quality is the silhouette score (S), which measures how similar a node is to its own cluster compared to other clusters (Rousseeuw 1987). The S score ranges from -1 to $+1$, where positive values indicate well-defined clusters. Specifically, a silhouette score exceeding 0.3 suggests a homogeneous network, 0.5 indicates a reasonable clustering structure, and 0.7 or higher reflects a highly reliable classification (Sabé et al. 2023, 3).

In addition to network structure, burst detection is used to analyze the temporal dynamics in scholarly communication by identifying sudden increases in the frequency of an event – such as the rapid citation of a document or the frequent occurrence of a keyword – over a specific period. Burst detection analyzes the frequency of specific events (e.g., keyword occurrences or citation counts) over time, looking for sharp, statistically significant increases compared to prior periods, which indicate a surge in attention toward a particular topic or work. CiteSpace quantifies burstiness using burst strength, which measures both the intensity and duration of a node's burst behavior within the network (CiteSpace 2025). Intensity refers to how high the frequency spikes during the burst, while duration reflects how long the burst maintains its elevated level. A higher burst strength indicates a more significant and prolonged surge in activity related to a term or paper, helping to track emerging research trends and identify influential works that experience a surge in academic attention.

4. Results

4.1. Research output and collaboration structure

4.1.1. Annual research output

Figure 2 illustrates the trend in annual research output on research integrity as derived from the merged dataset. The earliest publication in this field,

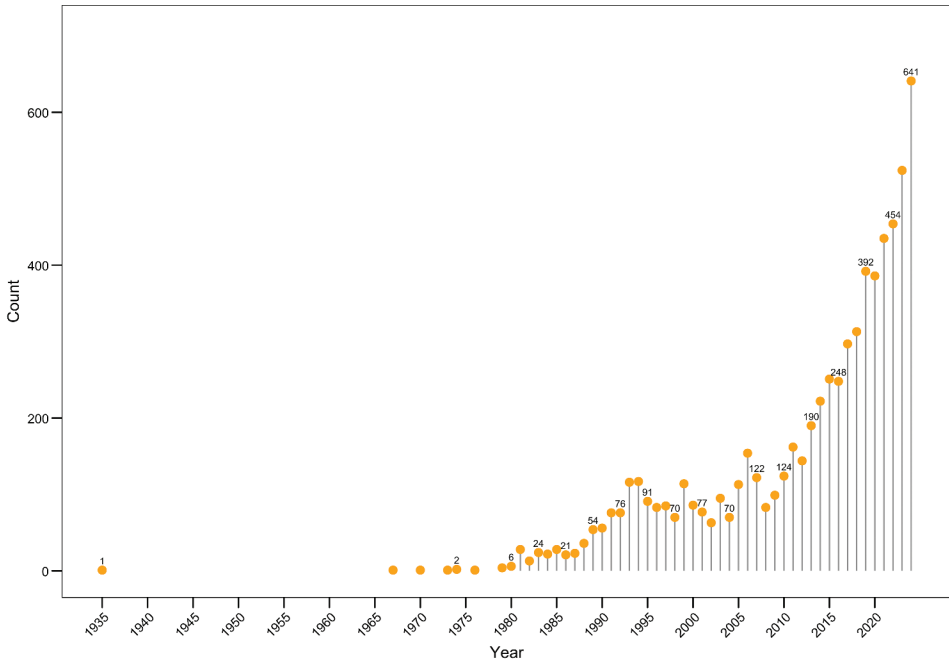


Figure 2. Trend of annual research output on research integrity.

which employed the term “scientific integrity,” appeared in 1935. This publication sought to defend William James’s scientific integrity, particularly in response to criticisms regarding his interest in psychical research (Dearborn 1935). The presence of this early record indicates that discussions surrounding research integrity had already emerging by 1935.

For nearly four decades following this initial contribution, academic output on research integrity remained limited. A gradual increase in publications was not observed until after 1980. By 2010, the annual scholarly output reached 124 publications, and this trend accelerated in subsequent years, culminating in 641 publications in 2024.

4.1.2. Co-authorship and journal clustering in research integrity studies

Figure 3 presents the clustered co-authorship and journal distribution networks in research integrity.

Figure 3(a) illustrates the co-authorship network, highlighting key authors and collaboration patterns. The largest cluster (#0), centered on “research integrity,” features Lex Bouter and Ana Marušić as leading figures, emphasizing their contributions to this field. In the “responsible research and innovation” cluster (#1), key contributors include David B. Resnik, Arturo Casadevall, and Hosseini Mohammad, and in the “scientific misconduct”

cluster (#8), Gert Helgesson and Björn Hofmann are prominent figures. The “human brain project” cluster (#4), led by Bernd Carsten Stahl and Inga Ullmann, explored ethical challenges in neuroscience.

Figure 3(b) presents the journal co-citation network, mapping the citation relationships among key journals in research integrity literature. The “research integrity” cluster (#0), centered around *BMJ*, *PLOS ONE*, *JAMA*, and *The Lancet*, occupies a pivotal position, reflecting the prominence of medical and multidisciplinary journals in integrity discussions. The “responsible research” cluster (#1), which includes *Science and Public Policy* and *Research Policy*, suggests a policy-oriented perspective on ethical research practices. The “publication ethics” cluster (#2), featuring *Nature*, *Science*, and *The New England Journal of Medicine (NEJM)*, underscores the role of high-impact journals in shaping peer review and editorial standards. Meanwhile, the “social responsibility” cluster (#3), represented by *The Journal of Business Ethics* and *Accountability in Research*, signals an intersection between research integrity and broader ethical considerations.

4.1.3. Global and institutional collaboration in research integrity studies

Figure 4 illustrates the global and institutional collaboration networks in research integrity, highlighting key contributing countries, institutional partnerships, and evolving thematic trends.

At the global level (see Figure 4(a)), the United States, the United Kingdom, Germany, and the Netherlands have traditionally been among the most influential contributors, maintaining extensive international collaborations. Since 2000, Asian countries, including China, Malaysia, Singapore, Japan, and South Korea, have been increasingly involved in discussions on research integrity, particularly in areas related to “paper mills” (cluster #5) and regional research integrity concerns (cluster #3). Scandinavian countries, such as Norway, Denmark, and Sweden, have exhibited cross-regional collaborations, with a primary focus on academic “dishonesty” (cluster #4). Meanwhile, Australia and New Zealand have placed a greater emphasis on topics related to “data integrity” (cluster #6).

At the institutional level (see Figure 4(b)), early contributions were predominantly from U.S. institutions, including the University of California, the University of Michigan, the National Institutes of Health (NIH), and the University of Washington. However, after 2010, research integrity discussions became increasingly centered in European institutions. Universities such as Vrije Universiteit Amsterdam, Utrecht University, Maastricht University, the University of Copenhagen, Leiden University, and Aarhus University have emerged as key players in this field. North American institutions tend to focus on topics such as “data integrity” (cluster #4), “informed consent” (cluster #2), and “journal publication” (cluster #2), whereas European institutions are more engaged in “neo-institutional theory” (cluster

#0) and responsible “research and innovation (RRI)” (cluster #3) in the context of research integrity. Over time, institutional collaborations have strengthened, fostering more cross-border research partnerships.

Expanding collaborations indicate that research integrity is evolving into a global, interdisciplinary field that integrates policy, technology, and governance.

4.2. Terminology and keyword and co-occurrence and burst detection analysis

4.2.1. Term co-occurrence analysis

The term co-occurrence network maps the conceptual evolution of research integrity, showing how its key ideas have been structured and interconnected over time (Cheng et al. 2018). The evolution of terminology in research integrity progressed from a scattered distribution to a highly interconnected network, displaying notable structural differences across various time periods (see Figure 5(a)). In the early stage (1935–1984), the term network was relatively sparse (see Figure 5(c)), lacking complex cluster interactions and showing weak connections between key terms.

Between 1985 and 1994, the term co-occurrence network showed increasing internal cohesion within specific clusters (see Figure 5(d)), reflecting the gradual structuring of knowledge in research integrity. For instance, within the “scientific misconduct” cluster (#1), “professional ethics” became increasingly connected to “data selection” and the “Helsinki Declaration,” signaling a shift from a primarily moralistic discussion of individual integrity to a broader ethical framework (Berglund and McNeill 1989). However, while the number of core terms increased, clusters remained relatively isolated, indicating that research integrity had not yet developed into a fully integrated conceptual system but was still characterized by independent yet emerging discussions.

Between 1995 and 2004, the term network in research integrity became more structured, forming a dual-core structure centered on “scientific misconduct” (cluster #1) and “research integrity” (cluster #0), while strengthening its connections with surrounding clusters. Within “scientific misconduct” (cluster #1), the strengthened connection between “ethics in science” and “professional ethics” reflects a growing refinement in the conceptualization of misconduct. During this period, research misconduct was no longer defined solely as individual violations but was increasingly framed as deviations from professional norms and academic standards. Its classification became more systematic, with clearer distinctions among different types of misconduct, such as data fabrication, plagiarism, and conflicts of interest (Lock 1996; Lynoe, Jacobsson, and Lundgren 1999; Parrish 1996). At the same time, “research integrity” (cluster #0) established stronger associations

CytoSpace, v. 6.4.R1 (64-bit) Advansee
February 2, 2025, 8:22:39 PM PST
Virus: O-*Saccharomyces cerevisiae* Analysis of Research Integrity/Analysis process/output
Timeline: 1930-2024 (Silo: Longest/No)
Selection Criteria: Top 63 per slice, LRF=+0.0, LBN=-0.0, LBH=-1.0, em=0
Network: AMR6, G=67 (classicalG=3110)
Nodes Labeled: 1.0%
Pruning: Pathfinder
Modularity Q=0.621
Weighted Mean Silhouette S=0.9459
Harmonic MeanQ+S=0.7739
Collapsed



CitSpace v. 5.6.R1 (64-bit) Advansoft
February 2, 2020, 8:32:31 PM GMT
Input: C:\Documents\Analysis of Research Integrity\Analysis process\out
View: Modularity Q=0.9719 Weighted Mean Silhouette S=0.9719
Detection Criteria: Top 60 per edge, LRF=0.5, L/N=5, L/U=11, e=0.0
Network N=247, E=116 (Density=0.0087)
Largest CCs: 113 (65%)
Nodes Labeled: 1.0%
Pruning: Pathfinder
Modularity Q=0.9719
Weighted Mean Silhouette S=0.9719
Harmonic Mean Q, S=0.9717



with “research ethics” (cluster #4) and “scientific fraud” (cluster #3), indicating that research integrity has expanded beyond ethical principles to address issues of misconduct and regulatory challenge (Fuchs and Davis Westervelt

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1996). Moreover, “conflict of interest” played a key role in bridging “scientific misconduct,” “editorial policies,” and “public health,” reflecting the extension of the research integrity discourse into institutional policies, regulatory frameworks, and specific disciplinary applications (Atlas 2003; Cossette 2004). Additionally, “data audit” (cluster #7) and “medical controversy” (cluster #12) established new conceptual connections with “peer review” and “data integrity” within the “research integrity” cluster (#0). This pattern highlighted a characteristic of research integrity during this period – its increasing entanglement with bioethics.

Compared to the previous period, the 2005–2014 term network exhibited stronger overall cohesion, with a significant increase in cross-cluster conceptual connections (see Figure 5(f)), leading to a more tightly interwoven conceptual structure. From a structural perspective, while some cross-cluster connections existed earlier, distinct subfields were still identifiable. However, between 2005 and 2014, the boundaries between clusters gradually blurred, and conceptual linkages significantly intensified, indicating a further consolidation of the research integrity knowledge system. This structural transformation suggested that research integrity evolved from a collection of relatively independent conceptual units into a more interconnected and integrated knowledge system. On the one hand, terms such as “conflict of interest,” “editorial policies,” “medical journals,” “research ethics

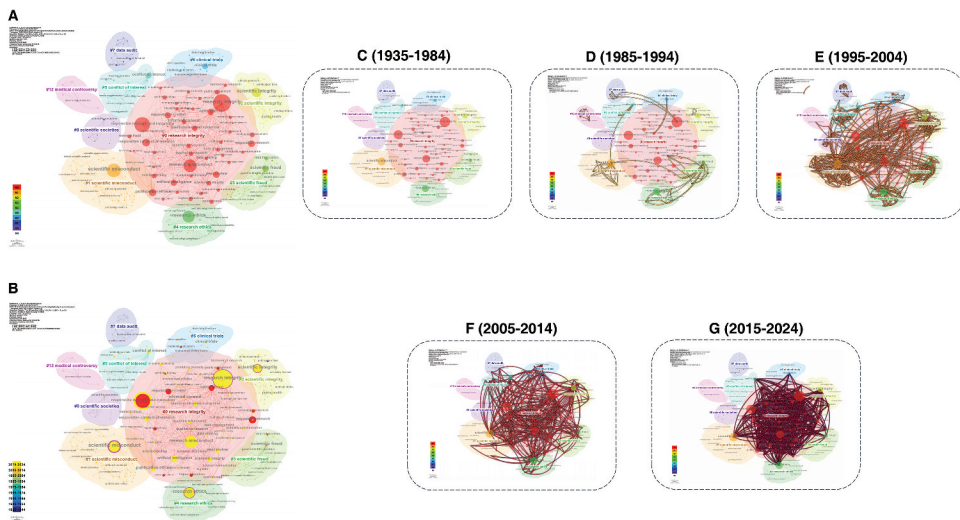


Figure 5. Term co-occurrence network analysis results. (A) Overview of the term co-occurrence network. (B) Hotspot detection results. (C) Term co-occurrence network from 1935 to 1984. (D) Term co-occurrence network from 1985 to 1994. (E) Term co-occurrence network from 1995 to 2004. (F) Term co-occurrence network from 2005 to 2014. (G) Term co-occurrence network from 2015 to 2024. Although the analysis was conducted in 10-year intervals, certain intervals (1935–1944, 1945–1954, 1955–1964, 1965–1974, 1975–1984) showed no changes in the network structure. For a more concise visualization, these slices are merged into a single figure.

committees,” and “data fabrication” served as bridging concepts across clusters, facilitating greater conceptual interaction and integration. On the other hand, terms like “duplicate publication,” “clinical trials,” and research ethics established stronger associations with core concepts such as research integrity, gradually increasing their influence within the conceptual system and becoming integral to the discourse on research integrity. In other words, research integrity was no longer a loosely connected domain composed of isolated discussions but had evolved into a more cohesive conceptual framework, where dialogue and coordination between different concepts became increasingly systematic.

From 2015 to 2024, the term network showed a high degree of integration, with nearly all core themes converging around research integrity at the center. The connections among the nodes became extremely dense, suggesting that research integrity had reached a stage of profound, multi-topic convergence. “Research integrity” (cluster #0), “scientific misconduct” (cluster #1), “scientific integrity” (cluster #2), “scientific fraud” (cluster #3), and “research ethics” (cluster #4) continued to serve as the principal pillars of the network; however, the number of links among them rose markedly, weaving these clusters into a closely interactive knowledge system. Meanwhile, “conflict of interest” (cluster #5), “clinical trials” (cluster #6), and “data audit” (cluster #7) remained strongly tied to the central topics but featured few, if any, internal links. This implied that these areas were no longer functioning as independent conceptual clusters. Instead, they appeared to funnel toward key concept clusters (e.g., “research integrity” (cluster #0) or “scientific misconduct” (cluster #1)), reflecting their conceptual integration within the broader discourse rather than serving as distinct domains. Within the research integrity cluster (#0), terms such as “open access,” “responsible research and innovation,” “open science,” “artificial intelligence,” and “machine learning” developed more connections with other terms, aligning closely with the hotspot detection results (see [Figure 5\(b\)](#)). This pattern indicates that emerging interdisciplinary concepts are becoming embedded within the broader research integrity discourse (Popkov and Barrett 2024; Tang and Elaine Eaton 2024).

4.2.2. Keyword co-occurrence and burst detection

Unlike term co-occurrence analysis, which constructs a knowledge map, analyzing keyword co-occurrence relationships (see [Figure 6\(a\)](#)) reveals core research themes and their interconnections (Mokhtarpour and Akbar Khasseh 2020). Burst detection (see [Table 1](#)) identifies keywords with sudden surges in usage over specific periods, indicating emerging research trends or rapidly rising topics in the field (Chen 2006). This approach not only facilitates the tracking of research topic evolution but also captures knowledge dynamics that may not be fully reflected in term-based networks.

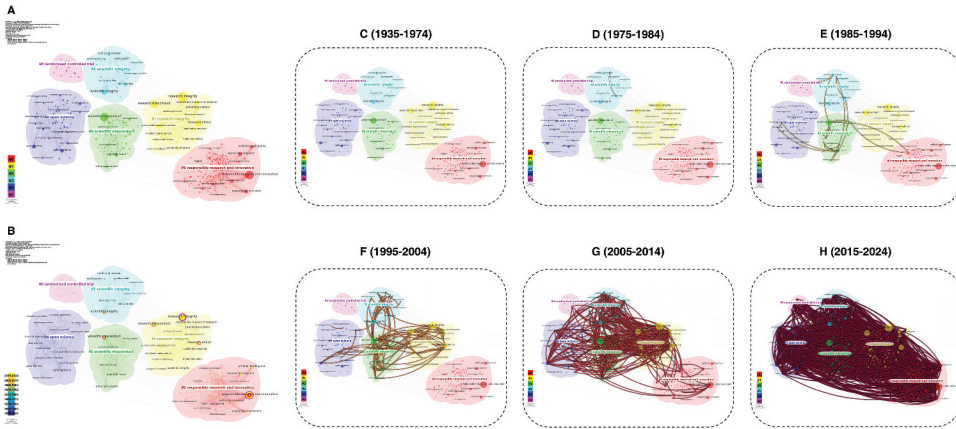


Figure 6. Keyword co-occurrence network analysis results. (A) Overview of the keyword co-occurrence network. (B) Hotspot detection results. (C) Keyword co-occurrence network from 1935–1974. (D) Keyword co-occurrence network from 1975–1984. (E) Keyword co-occurrence network from 1985–1994. (F) Keyword co-occurrence network from 1995–2004. (G) Keyword co-occurrence network from 2005–2014. (H) Keyword co-occurrence network from 2015–2024. Although the analysis was conducted in 10-year intervals, certain intervals (1935–1944, 1945–1954, 1955–1964, 1965–1974) showed no changes in the network structure. For a more concise visualization, these slices are merged into a single figure.

In the 1935–1974 co-occurrence network (see [Figure 6\(c\)](#)), the absence of substantial connections between keywords suggested that research integrity was not yet established as a cohesive research domain, and discussions remained fragmented across independent topics. The loosely connected structure indicated that research themes were not yet well-defined, with weak associations between topics, reflecting that research integrity was still an isolated and sporadically discussed topic.

Between 1975 and 1984 (see [Figure 6\(d\)](#)), keyword co-occurrence networks began to show initial signs of connectivity, particularly between “conflict of interest” and “clinical trials,” suggesting that discussions on research integrity were centering on specific ethical concerns. For instance, a pivotal article published in *NEJM* in 1982 systematically examined conflicts of interest, informed consent, and treatment preferences (Schafer 1982), sparking broader discussions in subsequent years (Cancer Research Campaign Working Party in Breast Conservation 1983; Dudley 1983). However, the network at this stage had not yet developed cross-cluster structures, implying that discussions on research integrity were still in their formative stage, with weak interconnections between topics.

Between 1985 and 1994 (see [Figure 6\(e\)](#)), the keyword co-occurrence network exhibited an increase in linkages, with stronger internal connections within clusters and an increasing number of cross-cluster associations.

Compared to the previous period, the links between “data sharing,” “scientific fraud,” and “data fabrication” became more prominent, signaling a shift in focus beyond traditional research ethics to encompass data management, publication standards, and research reproducibility (Block and Hartman 1989; Bross 1990). However, while these changes indicate an emerging structure, the overall network remained somewhat dispersed, lacking a centralized framework. Compared to earlier periods, the organization of the keyword network improved, suggesting a transition from isolated case studies to broader explorations of the research ecosystem, supporting scholarly and policy discussions (Gunsalus 1993; Sachs and Siegler 1993).

Between 1995 and 2004 (see Figure 6(f)), the keyword network in research integrity became more interconnected, with a significant increase in cross-cluster linkages, marking a transition toward greater thematic integration. The connection among “scientific misconduct” (cluster #2), “research integrity” (cluster #1) and “open science” (cluster #4) strengthened, and subtopics such as “scholarly communication,” “science policy,” “publication ethics,” and “peer review” exhibited stronger cross-cluster linkages. This indicates that discussions in this domain had expanded beyond ethical considerations into broader institutional and governance frameworks (Cronin 2001; Stern and Simes 1997). Additionally, keywords such as “scientific fraud,” and “duplicate publication” emerged as bridge terms linking different clusters, reflecting increasing scholarly attention to the specific manifestations of research misconduct. The discussion on research integrity was becoming more nuanced and granular.

From 2005 to 2014 (see Figure 6(g)), the keyword network became denser, with a marked increase in network complexity and interconnections between previously distinct research topics. “Scientific misconduct” (cluster #2) emerged as a central hub within multiple clusters, reinforcing its pivotal role in discussions on research integrity. “Open science” (cluster #4) became an emerging research focus, with its subtopics, such as “peer review” and “data management,” forming extensive connections with other areas (Caulfield, Harmon, and Joly 2012; Tenopir et al. 2011). This shift highlighted the growing academic interest in transparency, reproducibility, and data governance. Moreover, discussions on “research integrity” (cluster #1) established increasing connections with the “responsible research and innovation” cluster (#0), indicating that the governance of research misconduct was becoming more integrated with research policies, ethical governance, and societal responsibilities (Ann Lind 2005; Resnik, Peddada, and Brunson 2009).

Between 2015 and 2024 (see Figure 6(h)), the density of keyword co-occurrence reached its highest level, with significantly enhanced cross-domain linkages. Discussions on research integrity had become highly integrated and were progressively expanding into broader science policy debates. “Responsible research and innovation” (cluster #0) emerged as a central

theme, with its internal keywords, such as “citizen science” and “public engagement,” forming strong connections with other clusters, suggesting that research integrity had evolved from a focus on academic conduct to a broader scientific social responsibility discourse (Ciubotariu and Bosch 2022; Resnik, Elliott, and Miller 2015). The emergence of “artificial intelligence,” “big data,” and “science policy,” and their strong connections with keywords within the “research integrity” cluster (#1), highlight the ethical challenges posed by technological advancements (Moffatt and Hall 2024; Raman 2023). This indicates that discussions on research integrity have extended beyond academic settings, increasingly intersecting with societal and policy concerns. Furthermore, “open science” (cluster #4) continued to gain prominence in this period, with its internal keywords, such as “research assessment” and “predatory journal,” exhibiting increased connections, emphasizing that integrity challenges associated with open access and research transparency are becoming more significant (Gerashchenko 2022; Kharumnuid and Singh Deo 2022; Tang and Jia 2021). The sudden increase in inter-cluster connections further suggested that research integrity has become a focal point for cross-disciplinary collaboration and governance efforts.

Integrating burst detection data, keyword strength (see Table 1), and topic trend analysis (see Figure 6(b)), several key research integrity themes are expected to dominate future discussions, including the impact of artificial intelligence on research integrity, the deepening of open science initiatives, reforms in peer review systems, and the increasing role of public engagement. The intersection of AI and research integrity is anticipated to be a major area of focus. The burst detection of “machine learning” (2020, strength: 5.63) suggests that AI is increasingly influencing academic publishing and research data analysis. This trend has sparked concerns regarding automated manuscript generation, peer review bias, and algorithmic transparency. The burst detection of “synthetic biology” in 2013 (5.98) highlights ethical concerns in emerging research fields. Future intersections between AI and synthetic biology may introduce novel regulatory challenges. Additionally, the increasing burst strength of open science since 2015 (5.83) underscores the growing emphasis on transparency, data sharing, and reproducibility in scholarly research. When combined with the persistent burst of publication bias (2005, 5.44) and data fabrication (1995, 5.73), it becomes evident that issues such as data manipulation, selective reporting, and transparency in peer review remain critical challenges. The burst detection of public engagement (2013, 5.43) suggests that public participation in research is becoming increasingly important, contributing to enhanced risk assessment and ethical oversight in scientific investigations (Dorris 2024).

Table 1. Burst detection results for keywords.

Keywords	Year	Strength	Begin	End	1935-2024
scientific misconduct	1991	34.54	1991	2014	
scientific fraud	1991	17.27	1991	2014	
conflict of interest	1992	7.06	1992	2014	
informed consent	1993	5.58	1993	2014	
scientific integrity	1995	10.33	1995	2014	
research ethics	1995	4.93	1995	2014	
research misconduct	1996	10	2005	2014	
responsible conduct of research	1998	9.32	2005	2014	
publication ethics	1996	7.06	2005	2024	
synthetic biology	2013	5.98	2013	2024	
data fabrication	1995	5.73	2005	2024	
peer review	1995	5.49	2005	2014	
publication bias	2005	5.44	2005	2024	
emerging technologies	2012	4.73	2012	2024	
open science	2015	5.83	2015	2024	
machine learning	2020	5.63	2020	2024	
public engagement	2013	5.43	2015	2024	

4.3. Co-citation analysis and burst detection of references

To further elucidate the knowledge evolution of research integrity and identify potential emerging trends, a co-citation analysis of references was conducted (see [Figure 7\(a\)](#)). Similar to the co-occurrence network of terms, co-citation analysis reveals structural connections within the knowledge system. In the period between 1935 and 1984 (see [Figure 7\(c\)](#)), co-citation

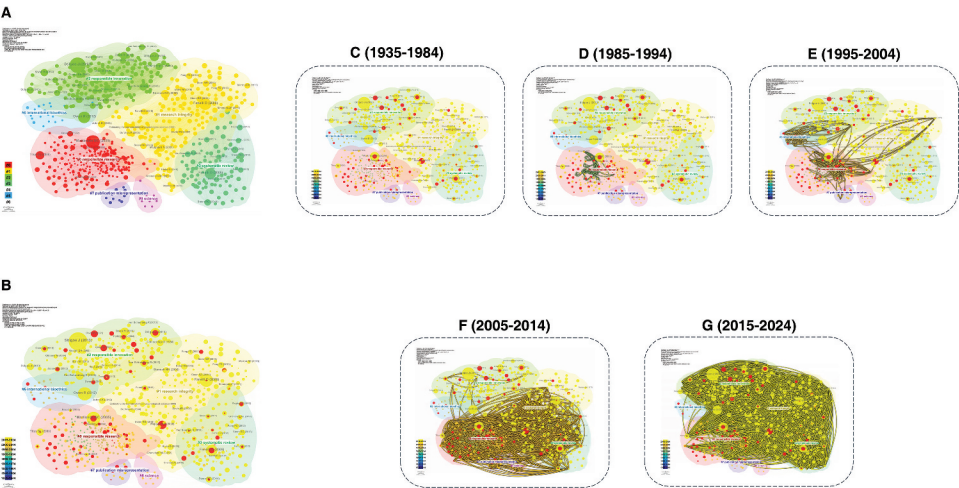
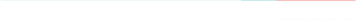


Figure 7. Co-citation network analysis of references. (A) Overview of the co-citation network. (B) Burst detection of co-cited references. (C) Co-citation network of references from 1935–1984. (D) Co-citation network of references from 1985–1994. (E) Co-citation network of references from 1995–2004. (F) Co-citation network of references from 2005–2014. (G) Co-citation network of references from 2015–2024. Although the analysis was conducted in 10-year intervals, certain intervals (1935–1944, 1945–1954, 1955–1964, 1965–1974, 1975–1984) showed no changes in the network structure. For a more concise visualization, these slices are merged into a single figure.

Table 2. Burst detection results for co-cited references.

Literature	Year	Strength	Begin	End	1935-2024
Broad and Wade 1982	1982	9.84	1985	2014	
Stewart and Feder 1987	1987	7.67	1987	2004	
Culliton 1983	1983	6.43	1985	2014	
Flanagin 1998	1998	9.08	1998	2014	
Kalichman and Friedman 1992	1992	8.12	1995	2014	
Nylenna et al. 1999	1999	6.83	1999	2014	
Gurudevan and Mower 1996	1996	5.43	1996	2014	
Friedman 1990	1990	5.18	1995	2014	
Rankin and Esteves 1997	1997	4.89	1997	2014	
Martinson, Anderson, and de Vries 2005	2005	44.46	2005	2024	
Titus, Wells, and Rhoades 2008	2008	15.3	2008	2024	
Fanelli 2009	2009	14.38	2009	2024	
Anderson et al. 2007	2007	13.86	2007	2024	
Steen 2011	2011	10.95	2011	2024	
Begley and Lee 2012	2012	9.94	2012	2024	
Wager and Williams 2011	2011	5.99	2011	2024	
Plemmons, Brody, and Kalichman 2006	2006	5.93	2006	2024	
Van Noorden 2011	2011	5.58	2011	2024	
Grieneisen and Zhang 2012	2012	5.55	2012	2024	
Fang and Casadevall 2011	2011	4.73	2011	2024	
Mejlgaard et al. 2020	2020	16.26	2020	2024	
Owen, von Schomberg, and Macnaghten 2021	2021	14	2021	2024	
Stilgoe, Owen, and Macnaghten 2013	2013	9.02	2015	2024	
Fanelli, Rodrigo, and Vincent 2015	2015	7.46	2015	2024	
Burget, Bardone, and Pedaste 2016	2017	7.41	2017	2024	
Owen et al. 2021	2021	6.54	2021	2024	
Pizzolato, Abdi, and Dierickx 2020	2020	5.66	2020	2024	
Brand and Blok 2019	2019	5.5	2019	2024	
Wiarda et al. 2021	2021	5.28	2021	2024	
Else and Van Noorden 2021	2021	5.28	2021	2024	

relationships among reference nodes were nearly absent, indicating weak intellectual influence across different research directions.

In the 1985–1994 co-citation analysis of references (see Figure 7(d)), the primary co-citation relationships were concentrated within the “responsible research” cluster (#0). This suggested that research within this domain began to consolidate, forming a more cohesive knowledge foundation and fostering stronger internal intellectual connections. However, due to the absence of cross-cluster linkages, studies during this period remained largely confined to their respective fields, lacking significant integration. Notably, burst detection of co-cited references (see Table 2) indicated that *Betrayers of the Truth: Fraud and Deceit in the Halls of Science* by Broad and Wade (1982) emerged as a highly cited work starting in 1985, becoming an influential reference during this period. The book critically challenged the prevailing assumption within the academic community that scientific errors could be swiftly corrected through rigorous peer review and experimental replication, calling into question traditional perceptions of research processes. This perspective gained widespread attention in discussions on research integrity, prompting a reassessment of the systemic causes of research misconduct.

In the 1995–2004 co-citation analysis of references (see Figure 7(e)), the overall structure became significantly more interconnected than in the

previous period, with the emergence of cross-cluster linkages, marking an increasing degree of knowledge integration in research integrity. Structurally, core themes such as “responsible research” (cluster #0), “research integrity” (cluster #1), “international bioethics” (cluster #5), and “publication misrepresentation” (cluster #7) not only developed more stable internal co-citation relationships but also established cross-cluster connections through bridging references, notably *The Sociology of Science: Theoretical and Empirical Investigations* by Merton (1973). Although this work was originally published in 1973, its influence on research integrity became more pronounced during 1995–2004, serving as a crucial theoretical foundation linking different sub-fields. As a seminal work in the sociology of science, it integrated sociological perspectives with institutional analyses of science, providing a theoretical basis for understanding deviant behaviors in science, academic norms, and ethical concerns in research practices. Furthermore, the increasing co-citation relationship between key references in the responsible research and “international bioethics” (cluster #5) clusters suggests that ethical considerations gained prominence in discussions on research integrity. This trend may be linked to the further development of biomedical research ethics frameworks in the late 1990s, including the expanded application of the *CIOMS Guidelines* and the *Belmont Report* in ethical oversight, reinforcing the role of ethics as an integral part of research integrity discussions (Idänpään-Heikkilä and Fluss 2003; Weijer 1999).

In the 2005–2014 co-citation analysis of references (see Figure 7(f)), the overall network structure became even more tightly interconnected, with a significant increase in cross-cluster co-citation relationships, reflecting further consolidation of the knowledge system in research integrity. The co-citation links between core themes such as “responsible research” (cluster #0) and “research integrity” (cluster #1) strengthened. In contrast, “international bioethics” (cluster #5) showed weaker connectivity, suggesting that research ethics discussions were becoming more autonomous. This shift may be associated with the further institutionalization of biomedical ethics as a distinct field during this period. Meanwhile, connections between “responsible research” (cluster #0) and “responsible innovation” (cluster #2) began to emerge, indicating an expansion of research integrity discussions into science policy and innovation governance, likely driven by policy initiatives and technological developments (Stahl 2013; Wickson and Carew 2014). Additionally, the “systematic review” (cluster #3) cluster demonstrated increased co-citation relationships, reflecting a growing emphasis on research reproducibility, data transparency, and evidence synthesis in the research integrity discourse. This trend may have been influenced by methodological advancements and heightened discussions surrounding the replication crisis (Mahood, Van Eerd, and Irvin 2014; Thayer et al. 2014). Notably, some references with high citation counts served as intellectual bridges

across multiple topics. These included empirical studies on data fabrication among scientists (Fanelli 2009; Martinson, Anderson, and de Vries 2005), conceptual clarifications of research integrity (Steneck 2006), and investigations into retraction causes (Fang, Steen, and Casadevall 2012). These bridging references contributed to a more complex knowledge network for research integrity.

In the 2015–2024 co-citation analysis of references (see Figure 7(g)), the overall structure exhibited a high degree of connectivity, with extensive cross-cluster co-citation relationships. Both older and recent references were increasingly co-cited, suggesting that the knowledge system of research integrity had reached a highly integrated stage. The interconnections among core themes such as “responsible research” (cluster #0), “research integrity” (cluster #1), “systematic review” (cluster #3), and “responsible innovation” (cluster #2) further strengthened, indicating that the field had transitioned from relatively independent research areas into a highly integrated and cohesive knowledge system. Compared to the 2005–2014 co-citation network, the importance of “responsible innovation” (cluster #2) increased further during this period, forming a tighter link with “responsible research” (cluster #0), aligning with the broader integration of “responsible research and innovation.” Additionally, key references with strong citation bursts played a crucial role in establishing knowledge linkages across multiple clusters. These references focused on retraction investigations and their increasing causes (Grieneisen and Zhang 2012; Steen, Casadevall, and Fang 2013; Van Noorden 2011), different features of the RRI concept (Owen, Macnaghten, and Stilgoe 2012), and the improvement of research culture, publication practices, and incentive structures (Begley and Lee 2012). The strengthening of co-citation relationships among core references and the intensified cross-cluster knowledge exchange reflect the growing recognition of research integrity as a critical cross-sectoral research domain.

Burst detection of co-cited references (see Table 2) and hotspot analysis (see Figure 7(b)) revealed that foundational works, such as *Betrayers of the Truth: Fraud and Deceit in the Halls of Science* (Broad and Wade 1982) and the study by Martinson, Anderson, and de Vries (2005), played a pivotal role in shaping discussions on research integrity. After 2010, issues related to retractions became a central focus, with numerous burst references discussing various aspects of this topic (Fang and Casadevall 2011; Grieneisen and Zhang 2012; Steen 2011; Van Noorden 2011; Wager and Williams 2011). Meanwhile, discussions on responsible research and innovation produced several high-burst references (Brand and Blok 2019; Burget, Bardone, and Pedaste 2016; Owen et al. 2021; Owen, von Schomberg, and Macnaghten 2021; Stilgoe, Owen, and Macnaghten 2013; Wiarda et al. 2021). Other isolated burst references suggest emerging areas of interest, such as research culture, gender and research integrity (Fanelli, Rodrigo, and Vincent 2015),

paper mills (Else and Van Noorden 2021), and access to research integrity education resources (Mejlgaard et al. 2020; Pizzolato, Abdi, and Dierickx 2020). These trends reflect the expanding multidimensional trajectory of research integrity studies.

5. Discussion and conclusion

This study provides a comprehensive scientometric analysis of research integrity, revealing its conceptual evolution, structural trends, and emerging research frontiers.

5.1. Research output and collaboration structure

The rapid expansion of research integrity discourse, particularly post-2010, reflects its growing interdisciplinarity. Co-authorship and journal networks highlight key figures like Lex Bouter, Ana Marušić, David B. Resnik, and Hosseini Mohammad alongside the central role of medical journals (*BMJ*, *JAMA*, *The Lancet*) and policy-oriented venues (*Science and Public Policy*, *Research Policy*). Multidisciplinary journals (*Nature*, *Science*) drive discussions on publication ethics, and *The Journal of Business Ethics* and *Accountability in Research* contributes to the broader research ethics discourse. Research has shifted from U.S. dominance to increased European participation, particularly at the institutional level. Asian contributions to topics such as paper mills and research transparency have also grown. However, this finding has not been explicitly examined in previous literature.

Overall, this study addresses **RQ1** by analyzing the distribution patterns of major journals, authors, and countries across different domains of research integrity using clustering methods. Compared to the findings of Nguyen and Tuamsuk (2024), this approach provides a more refined analytical framework and offers new insights into collaboration patterns within the field of research integrity.

5.2. Evolution of research topics: terminology and keywords

While previous studies have reviewed specific aspects of research integrity, such as empirical research (Aubert Bonn and Pinxten 2019), research culture (Vidak et al. 2021), responsible research (Tijdink et al. 2021), and academic integrity (Maral 2024; Marques, Reis, and Gomes 2019), they have largely overlooked a holistic perspective on the production and circulation of knowledge in this domain. This study adopts a scientometric approach to examine the evolution of research integrity discourse.

The term co-occurrence network, which reflects conceptual changes, reveals a structural shift in research integrity discourse. Initially, discussions

were fragmented (1935–1984), but by 2015–2024, research integrity had evolved into a highly integrated conceptual system. Early debates focused on isolated ethical concerns. Between 1995 and 2004, a dual-core structure emerged around scientific misconduct and research integrity, increasingly linking with research ethics and academic governance. In later periods, the rise of open science, responsible research and innovation, and artificial intelligence in research integrity discussions highlighted growing intersections with policy, technological ethics, and public engagement. This increasing integration suggests that research integrity has matured into a clearly delineated field of study, with a interconnected knowledge system that continues to evolve.

Notably, this study observed a shift from conceptual fragmentation to consolidation in research integrity studies, suggesting the field is moving toward consensus-building. Prior studies indicate that conceptual core convergence within a field is associated with academic consensus (Shwed and Bearman 2010), whereas core churn is linked to innovation (Kedrick, Levitskaya, and Funk 2024). The increasing concentration of terms suggests a strengthening academic consensus, with researchers gradually coalescing around a shared understanding of research integrity's definition, theoretical framework, and core challenges. This trend may be driven by the global promotion of research integrity policies, such as the *Singapore Statement* and the *Hong Kong Principles* (Moher et al. 2020; Resnik and Shamoo 2011), as well as heightened academic attention to misconduct. The standardization of terminology contributes to the consolidation of research integrity as a systematic research domain rather than a collection of isolated discussions.

Keyword co-occurrence analysis further underscores the shifting research priorities over time. While scientific misconduct and conflicts of interest dominated early discussions, the 2010s saw growing attention to publication ethics, peer review, and systematic review, reflecting concerns over reproducibility and research credibility. The 2020s have introduced new challenges, including the emergence of machine learning (such as large language models like ChatGPT) and synthetic biology (e.g., increasing ethical and integrity concerns surrounding gene editing and reproducibility, which have become prominent since 2013). These developments raise novel ethical and regulatory considerations regarding research integrity. These findings address **RQ2**, offering insights into the evolution of key topics in research integrity (Suchikova and Tsybuliak 2024; Von Reyn, Amritava, and Hogle 2013).

5.3. Evolution of knowledge structure and co-citation network of references

Previous studies (Ali and Aboelmaged 2020; Mahmud and Ali 2021; Patra and Kumar Das 2019) lacked a co-citation network analysis of references in

the field of research integrity, limiting their ability to systematically identify structural knowledge evolution, track the diffusion of research topics, and predict emerging frontiers in the field. This study fills this gap by revealing the evolutionary trajectory of the research integrity knowledge network, characterized by a “problem-triggered, theory-integrated, and institutionally diffused” pattern.

Before the 1980s, the sparse co-citation relationships indicated that discussions on research integrity had not yet formed an independent research domain and were largely embedded within traditional disciplines such as medical ethics and the philosophy of science. After 1985, the citation burst of *Betrayers of the Truth: Fraud and Deceit in the Halls of Science* (Broad and Wade 1982), a critical work on scientific fraud, marked a turning point in the field. This shift was reflected in the co-citation network by the initial formation of the “responsible research” core cluster, signaling the transition of research integrity from fragmented discussions to an institutionalized topic.

From the mid-1990s to the early 2000s, the network structure changes highlight the pivotal role of theoretical frameworks in knowledge integration. Merton’s (1973) sociology of science emerged as a crucial intellectual bridge during this period, providing a shared theoretical foundation that facilitated dialogue across different subfields. Entering the twenty-first century, the increasing density of the knowledge network and the rise of cross-disciplinary linkages indicated that research integrity had fully entered the stage of systemic governance. Visualization analysis revealed the emergence of new clusters, such as “responsible innovation” and “systematic review,” reflecting a shift in research integrity discourse from traditional compliance-based control to anticipatory governance.

This transformation has expanded research focus to include technology ethics, open science, and broader governance issues. Simultaneously, the escalation of real-world challenges such as retraction mechanisms and paper mills has resulted in sustained high-intensity citation clusters in the co-citation network, underscoring the research community’s increasing demand for innovative governance tools. This evolutionary trajectory suggests that research integrity is not merely a discussion of norms and ethics but is embedded in institutional development and governance innovations within the research ecosystem. These findings partially address RQ3.

5.4. Potential future research hotspots

Based on keyword burst detection and co-citation analysis, future research on research integrity is likely to focus on the following areas, which also address RQ3.

- (1) **Artificial Intelligence and Research Integrity:** The burst detection of “machine learning” in the keyword co-occurrence analysis suggests that

- AI's role in research integrity is becoming a focal point. The discourse is shifting from traditional misconduct detection toward concerns over algorithmic bias, automation ethics, and AI's impact on scientific publishing.
- (2) **Open Science, Transparency, and Publication Ethics:** The bursts of “open science,” “data fabrication,” and “publication ethics” in keyword co-occurrence, along with the co-citation bursts of works addressing paper mills and retractions, underscore the sustained focus on challenges in scholarly publishing. The growing co-citation network around retraction patterns and predatory publishing suggests that discussions on research credibility and editorial integrity will continue to expand.
 - (3) **Institutional and Policy Frameworks:** The burst detection of “public engagement” in keyword co-occurrence, combined with the co-citation bursts of literature on responsible research and innovation, indicates that research integrity is increasingly being embedded within a broader institutional and policy-oriented framework. This shift reflects a transition from reactive measures (misconduct investigations) to proactive governance, integrating integrity principles into funding agencies, institutional policies, and regulatory frameworks.
 - (4) **Ethical Challenges in Emerging Scientific Fields:** The simultaneous bursts of “synthetic biology” and “emerging technologies” in keyword co-occurrence suggest that discussions on research integrity are expanding into fields characterized by rapid technological advancements, high uncertainty, and evolving ethical dilemmas.

6. Limitations

Despite providing a comprehensive scientometric analysis of research integrity, this study has several limitations that should be acknowledged.

- (1) The exclusion of editorials and commentaries may have led to the omission of influential perspectives that could contribute to a deeper understanding of research integrity discourse. These sources often reflect expert opinions and policy discussions that may not be captured in standard research articles.
- (2) The results of scientometric analyses are highly sensitive to search strategies. As this study could not incorporate an exhaustive set of search queries, some key literature may have been unintentionally omitted. This limitation underscores the inherent trade-off between search breadth and data manageability in scientometric studies.
- (3) The thematic analysis relies on keyword and term-based clustering, which may, in some cases, reflect the adoption of specific terminology rather than substantive shifts in research topics.

Notes

1. Due to missing metadata (such as references and keywords) in some documents, certain records were excluded from the term and keyword co-occurrence analysis, as well as from the co-citation analysis of references. The excluded documents can be identified in the upper left corner of the figures.
2. Co-Citation Network, nodes represent cited references, and “By Citation” refers to the total number of times a reference has been cited in all sampled papers. In the Author Co-Occurrence Network, nodes represent authors, and “By Citation” is calculated as the total number of citations received by all publications authored by a given individual. In the Country Co-Occurrence Network, nodes correspond to countries, with “By Citation” reflecting the cumulative number of citations received by all papers published by authors affiliated with that country.
3. In the Term and Keyword Co-Occurrence Network, nodes represent terms or keywords, with the “By Citation” attribute indicating the total number of citations received by the papers in which the given keyword or term appears. It is important to note that the keywords used in this co-occurrence network are those assigned by the authors within their articles, not those allocated by the literature database. To clarify how these terms are extracted, the process involves POS (Part of Speech) tagging, a linguistic term that classifies words based on their grammatical features, including syntactic functions and morphological changes, while also considering their lexical meaning. The POS tagging technique employed by CiteSpace is the Stanford NLP Group Part-of-Speech tagger. This tool is used to extract noun phrases (Noun Phrases) from sections such as the “Title,” “Abstract,” “Author Keywords,” and “Keywords Plus” of the articles. Noun phrases, typically consisting of one or more nouns, are key terms that represent the main concepts and topics in the research, making them central to the co-occurrence network construction (Li and Chen 2022, 229).

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Data availability statement

The data that support the findings of this study are openly available in Open Science Framework (OSF) at <http://doi.org/10.17605/OSF.IO/YKWPU>.

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